

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** questions.

1 Solve

(a) $\lim_{x \rightarrow 0} \left[\frac{2 - \sqrt{x+4}}{x(x+1)} \right]$

[4 marks]

(b) $\lim_{x \rightarrow 2^+} \frac{|4-2x|}{8-2x-x^2}$

[5 marks]

2 (a) Find the derivative of $f(x) = \frac{1}{1-x}$ using the first principle.

[4 marks]

(b) Given $y = \frac{\sin x}{1 + \cos x}$. Find $\frac{dy}{dx}$. Hence, find the value of $\frac{dy}{dx}$ when $x = 0$.

[5 marks]

3 Find the coordinates of the stationary points on the curve $f(x) = x^4 - 4x^3$ and determine their nature.

[7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1 Given $z = -3 + 4i$ and $zw = 14 - 2i$. Find

(a) w in the form of $a + bi$, where a and b are real numbers.

[3 marks]

(b) argument of w .

[2 marks]

2 Solve

(a) $2^{2x} - 9(2^{x-2}) + \frac{1}{2} = 0$

[6 marks]

(b) $\frac{-15-5x}{x-2} \geq x+3$

[5 marks]

3 (a) Given

$$f(x) = \begin{cases} -4x-1, & -4 \leq x < 0 \\ e^{2x-1}, & x \geq 0 \end{cases}$$

(i) Sketch the graph of $f(x)$

[3 marks]

(ii) State the range of $f(x)$.

[1 mark]

(iii) Evaluate $(f \circ f)(-1)$.

[3 marks]

(b) The functions g and f are given by

$$g(x) = 4 - 3x \text{ and } f(x) = x^2 + ax + b$$

where a and b are non-zero constants. Given that $(f \circ g)(2) = -5$ and $(g \circ f)(2) = -29$, find the value of a and b .

[5 marks]

4 Given

$$f(x) = \frac{1}{2}\sqrt{x-4}, \quad x \geq 5$$

(a) Find range of $f(x)$.

[1 mark]

(b) Determine $f^{-1}(x)$ and state its domain and range.

[4 marks]

(c) Hence, sketch the graph of $f(x)$ and $f^{-1}(x)$ on the same axes.

[3 marks]

5 (a) Find $\lim_{x \rightarrow \infty} \left(\frac{3x^2 + 7x - 1}{x^2 + 5} \right)^{\frac{1}{2}}$.

[4 marks]

(b) The function f is defined by

$$f(x) = \begin{cases} \frac{4}{\sqrt{4-x}}, & x < 0 \\ \sqrt{2}, & x = 0 \\ \frac{x}{\sqrt{1+x-1}}, & x > 0 \end{cases}$$

(i) Show that $\lim_{x \rightarrow 0} f(x)$ exist.

[7 marks]

(ii) Determine whether f is continuous at $x = 0$.

[3 marks]

6 (a) The equation of the curve is given by $e^y = \frac{x^2+3}{x-1}$, $x > 1$.

(i) Show that $\frac{dy}{dx} = \frac{(x-3)(x+1)}{(x^2+3)(x-1)}$.

[5 marks]

(ii) Find the coordinates when $\frac{dy}{dx} = 0$.

[3 marks]

(b) Given the parametric equations

$$x = \sec \theta \text{ and } y = \ln(1 + \cos 2\theta)$$

(i) Show that $\frac{dy}{dx} = -2 \cos \theta$.

[4 marks]

(ii) Hence, find $\frac{d^2y}{dx^2}$ when $\theta = \frac{\pi}{3}$.

[4 marks]

7 A closed, right circular cylinder of base r cm and height h cm has a volume of $54\pi \text{ cm}^3$.

(a) Show that S , the total surface area of the cylinder is given by $S = \frac{108\pi}{r} + 2\pi r^2$.

[4 marks]

(b) Hence, find the value of r which makes the surface area minimum.

[5 marks]

END OF QUESTION PAPER

Answers

Section A

- 1 (a) $-\frac{1}{4}$ (b) $-\frac{1}{3}$
- 2 (a) $\frac{1}{(1-x)^2}$ (b) $\frac{dy}{dx} = \frac{1}{1+\cos x}, \frac{1}{2}$
- 3 Stationary points: $(0,0)$, $(3,-27)$, Minimum point: $(3,-27)$, Inflection point: $(0,0)$

Section B

- 1 (a) $w = -2 - 2i$ (b) $\arg(w) = -\frac{3}{4}\pi$
- 2 (a) $x = -2, x = 1$ (b) $\{x : x < 2\}$
- 3 (a) (i) DIY
(ii) $R_f : [-1, \infty)$
(iii) $e^6 - 1$ (b) $a = 4, b = -1$
- 4 (a) $R_f : \left[\frac{1}{2}, \infty\right)$ (b) $f^{-1}(x) = 4x^2 + 4, D_{f^{-1}} : \left[\frac{1}{2}, \infty\right), R_{f^{-1}} : [5, \infty)$
- (c) DIY
- 5 (a) $\sqrt{3}$ (b) (i) Shown
(ii) $f(x)$ is not continuous at $x = 0$
- 6 (a) (i) Shown
(ii) $(3, \ln 6)$ (b) (i) Shown
(ii) $\frac{d^2y}{dx^2} = \frac{1}{2}$
- 7 (a) Shown (b) $r = 3\text{cm}$